

Sydney Girls High School



2004 Assessment Task 2

MATHEMATICS

Extension 2

= Year 12

Time allowed - 75 minutes

Topic: Complex Numbers

Instructions

NAME Sarah Feng

- Attempt all four questions.
- Questions are NOT of equal value.
- Total Marks = 75
- All necessary working should be shown in every question.
- Marks may be deducted for careless or badly arranged work.
- Write on one side of the paper only.

Question One (16 marks)

1. If $z = 3 + 4i$, $w = 3 - 2i$:
 - a) Find $z + w$ [1]
 - b) Find $z - w$ [1]
 - c) Find zw [1]
 - d) Find $\frac{z}{w}$ [1]
 - e) Find α given $\alpha^2 = z$ [3]
2. If $(1 + 2i)$ is a root of the quadratic equation $x^2 + bx + c = 0$ where b and c are real. Find b and c [2]
3. a) Solve the equation $z^7 = 1$ over the complex field giving your answers in mod/arg form [2]
- b) Given w is the root with the smallest positive argument show that $1 + w + w^2 + w^3 + w^4 + w^5 + w^6 = 0$ [2]
- c) Show that $\alpha = w + w^2 + w^4$ and $\beta = w^3 + w^5 + w^6$ are the roots of the equation $z^2 + z + 2 = 0$ [3]

Question Two (22 marks)

1. Draw sketch graphs of the following loci
 - a) $|z - 2 + 3i| = 4$ [2]
 - b) $|z| < 4$ and $\text{Re}(z) > -2$ [2]
 - c) $\arg(z - i) - \arg(z + i) = \frac{\pi}{2}$ [2]
 - d) $\text{Im}(z^2) = 4$ [2]
 - e) $1 \leq \text{Re}(z) \leq 2$ and $|\arg(z)| \leq \frac{\pi}{3}$ [2]
2. a) Sketch the locus of the complex number z where $|z - w| = \sqrt{5}$ given that $w = \frac{7 + 4i}{3 - 2i}$ [3]
- b) Use your diagram to find maximum $|z|$ and maximum $\arg(z)$ [5]
3. Let A be the complex number $(1 + i)$, B the complex number $(0 + i)$.
 A is rotated through π to A^* . B is rotated through $\frac{\pi}{2}$ to B^* .
 - a) Find the complex numbers representing A^* and B^* [2]
 - b) Hence describe the figure ABB^*A^* [2]

Question Three (16 marks)

1. Solve the following for x and y (both real) [3]

$$\frac{x+iy}{2+i} = 6-i$$

2. Find the Cartesian equation of the curve represented by [4]

$$|z|^2 - 2z - 2\bar{z} = 5$$

3. Given that $w = \frac{z+4i}{z-2}$ find the locus of w if w is purely real [5]

4. Solve the following for z , $\bar{z} + \frac{1}{z} = 2$ [4]

Question Four (21 marks)

1. Given $z = 8\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ and $w = 4\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$ find:

a) zw expressing your answer in the form $x + iy$ [2]

b) $\frac{z}{w}$ expressing your answer in the form $x + iy$ [2]

2. Simplify $(1 + \sqrt{3}i)^9$ [2]

3. By putting $z = (\cos\theta + i\sin\theta)$ and using De Moivre's Theorem

a) Express $\cos 4\theta$ in terms of powers of $\cos\theta$ and $\sin\theta$ [2]

b) Express $\sin 4\theta$ in terms of powers of $\cos\theta$ and $\sin\theta$ [1]

c) Express $\tan 4\theta$ in terms of powers of $\tan\theta$ [2]

d) Use your result from a) to find $\cos 4\theta$ purely in terms of powers of $\cos\theta$ [2]

e) Using your result from part d) and by putting $x = \cos\theta$, solve the equation $8x^4 - 8x^2 + 1 = 0$ [4]

f) Hence show that $\cos\frac{\pi}{8} + \cos\frac{3\pi}{8} + \cos\frac{5\pi}{8} + \cos\frac{7\pi}{8} = 0$ and [2]

g) Also that $\left(\cos\frac{\pi}{8}\right)\left(\cos\frac{3\pi}{8}\right)\left(\cos\frac{5\pi}{8}\right)\left(\cos\frac{7\pi}{8}\right) = \frac{1}{8}$ [2]

Extn 1
Solution, Task 2 2004

Q1. 1. $z = 3+4i$, $w = 3-2i$

a) $z+w = 3+4i + 3-2i = 6+2i$ # ①

b) $z-w = (3+4i) - (3-2i) = 6i$ # ①

c) $zw = (3+4i)(3-2i) = 9+8-6i+12i = 17+6i$ # ①

d) $\frac{z}{w} = \frac{3+4i}{3-2i} \cdot \frac{3+2i}{3+2i} = \frac{9-8+12i+6i}{13} = \frac{1}{13}(1+18i)$ # ①

2) let $z = a+bi$

$a^2-b^2+2abi = 3+4i$

$a^2-b^2 = 3$ ①, $2ab = 4$

$(a^2+b^2)^2 = (a^2-b^2)^2 + 4a^2b^2$

$= 25$

$a^2+b^2 = 5$

$2a^2 = 8$

$a = 2$ or $a = -2$

$b = 1$ or $b = -1$

$i.e. \pm (2+i)$ # ③

$\sqrt{5} \text{ cis}$

2. $(1+2i)$ is a root $\therefore (1-2i)$ is a root

$\alpha + \beta = 2$, $\alpha\beta = 5 \Rightarrow x^2 - (\alpha+\beta)x + \alpha\beta = 0$

$\therefore b = -2$, $c = 5$ # ②

3. a) $z^7 = 1$

$z^7 = \text{cis } 0$

$z^7 = \text{cis } (0+2\pi k)$ $k=0, 1, \dots, 7$

take $\sqrt[7]{}$ and apply De Moivre Th.

$z = \text{cis } \frac{2\pi k}{7}$

$z_0 = \text{cis } 0 = 1$, $z_1 = \text{cis } \frac{2\pi}{7}$, $z_2 = \text{cis } \frac{4\pi}{7}$, $z_3 = \text{cis } \frac{6\pi}{7}$ #

$z_4 = \text{cis } \frac{8\pi}{7}$, $z_5 = \text{cis } \frac{10\pi}{7}$, $z_6 = \text{cis } \frac{12\pi}{7}$ ②

b) $1+w+w^2+w^3+w^4+w^5+w^6$

Geometric $a=1$, $r=w$, $n=7$ and

its sum is given by

$S_n = \frac{a(r^n-1)}{r-1}$

$= \frac{1(w^7-1)}{w-1}$

$= 0$ since $w^7 = 1$ # ②

c) $\alpha + \beta = w + w^2 + w^3 + w^4 + w^5 + w^6$ OK sum of roots etc

$= (w + w^2 + w^3 + w^4 + w^5 + w^6 + 1) - 1$

$= -1$

$\alpha\beta = (w + w^2 + w^3)(w^4 + w^5 + w^6)$

$= w^6 + w^7 + w^8 + w^7 + w^8 + w^9 + w^8 + w^9 + w^{10}$

$= \underline{w^6} + \underline{w^7} + \underline{1} + \underline{w^5} + \underline{1} + \underline{w} + \underline{1} + \underline{w^4} + \underline{w^3}$

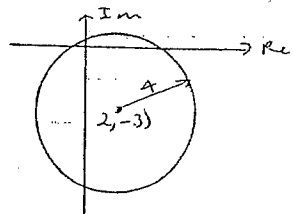
$= 2$

Quadratic of form

$x^2 - (\alpha + \beta)x + \alpha\beta = 0$

$x^2 + x + 2 = 0$ # ③

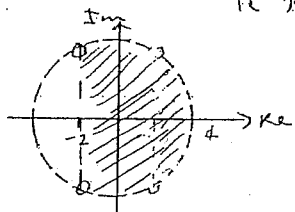
2) 1) a) $|z - 2 + 3i| = 4$
circle centre $(2, -3)$ $r = 4$



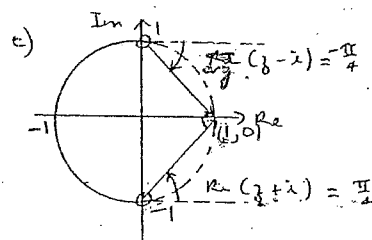
may have max
centre incorrect

②

b) $|z| < 4$, $\text{Re}(z) > -2$
ie $x > -2$



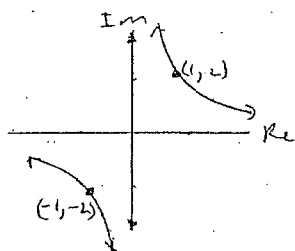
②



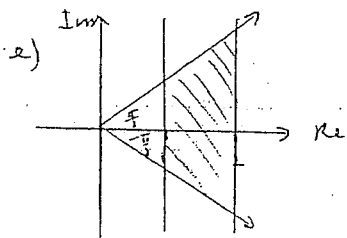
Test $(1, 0)$ RHS
 $\text{Re}(z - i) - \text{Re}(z + i)$
 $= -\frac{\pi}{4} - \frac{\pi}{4}$
 $= -\frac{\pi}{2}$
 $\therefore$ LHS of circle

②

d) $\text{Im}(z^2) = 4$
ie $\text{Im}[(x + iy)^2] = 4$
 $\text{Im}(x^2 - y^2 + 2xyi) = 4$
 $\therefore 2xy = 4$
 $xy = 2$



②



$1 \leq \text{Re}(z) \leq 2$
 $|\arg(z)| \leq \frac{\pi}{3}$

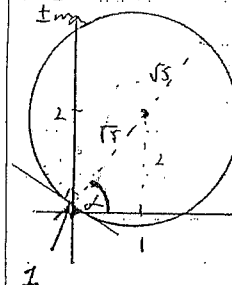
②

1 mark for top half on

Q2 Cont

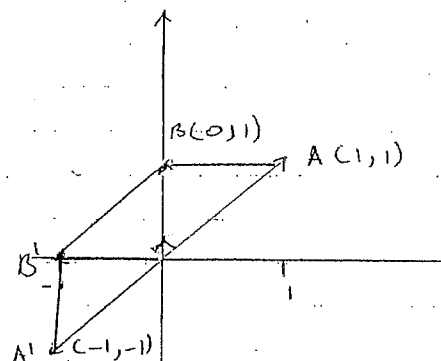
2) a) $|z - w| = \sqrt{5}$, $w = \frac{7+4i}{3-2i} \times \frac{3+2i}{3+2i}$

③ ie $|z - (1+2i)| = \sqrt{5}$
circle centre $(1, 2)$ $r = \sqrt{5}$
 $= \frac{21-8+12i+14i}{13}$
 $= 1 + 2i$



b) $\text{Max}|z| = 2\sqrt{5}$ ②

$\text{Max arg}(z) = \frac{\pi}{4} + \tan^{-1} 2$
or just less
 $\approx 153^\circ 26'$ ③



a) Rotate A through π is multiply by i^2
 $(1+i)i^2 = i^2 + i^3$
 $= -1 - i$ (A') ①

Rotate B through $\frac{\pi}{2}$ is multiply by i
 $(0+i)i = (-1+0)$ ①

b) $AB B' A'$ is an isosceles trapezium ②

Question Three

1. $\frac{z+iy}{z+i} = 6-i$

$$z+iy = (z+i)(6-i)$$

$$z+iy = 12+6i-2i-i^2$$

$$z+iy = 13+4i$$

$$\therefore x=13, y=4.$$

2. $|z|^2 - 2z - 2\bar{z} = 5$

$$|z|^2 - 2(z+\bar{z}) = 5$$

$$\text{Put } z = x+iy.$$

$$\text{then } x^2+y^2 - 2(2x) = 5$$

$$x^2 - 4x + 4 + y^2 = 9$$

$$4 \quad (x-2)^2 + y^2 = 3^2$$

Circle: Centre (2,0) Radius 3.

3. $w = \frac{z+4i}{z-2} = \frac{x+iy+4i}{x+iy-2} = \frac{x+i(y+4)}{(x-2)+iy}$

$$w = \frac{x+i(y+4)}{(x-2)+iy} \cdot \frac{(x-2)-iy}{(x-2)-iy} = \frac{x(x-2) - ixy + i(y+4)(x-2)}{(x-2)^2 + y^2}$$

$$w = \frac{x^2 - 2x - ixy + ixy - i^2y + i4x - 8i + y(y+4)}{(x-2)^2 + y^2}$$

w is purely Real $\therefore$ Imaginary Part is 0.

$$\therefore -2y + 4x - 8 = 0$$

$$2x - y = 4 \Rightarrow y = 2x - 4$$

excluding (2,0).

4. $\bar{z} + \frac{1}{z} = 2$

$$\therefore \bar{z} \cdot z + 1 = 2z$$

But $\bar{z} \cdot z$ is purely real

and $\bar{z} \cdot z + 1$ is purely real

$\therefore 2z$ is purely real.

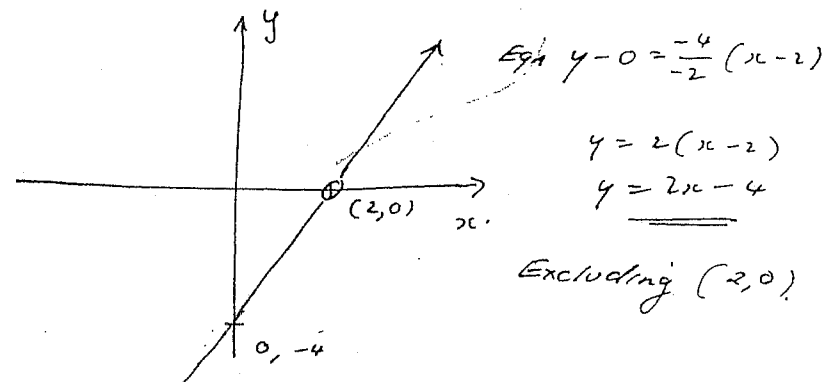
Hence: $\bar{z}^2 + 1 = 2z \Rightarrow \bar{z}^2 - 2z + 1 = 0$

$$(\bar{z}-1)^2 = 0$$

$$\therefore \underline{\underline{\bar{z} = 1}}$$

3. $w = \frac{z+4i}{z-2}$ w is purely real.
then $\arg w = 0, \pi$.

$$\text{i.e. } \arg\left(\frac{z+4i}{z-2}\right) = 0, \pi \Rightarrow \arg(z+4i) - \arg(z-2) = 0, \pi$$



Question Four

$$1. \quad z = 8 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$z = 8 \operatorname{cis} \frac{\pi}{3}$$

OR

$$z = 4 + 4\sqrt{3}i$$

$$w = 4 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

$$w = 4 \operatorname{cis} \frac{\pi}{6}$$

$$w = 2\sqrt{3} + 2i$$

$$2. \quad a) \quad zw = \left(8 \operatorname{cis} \frac{\pi}{3} \right) \left(4 \operatorname{cis} \frac{\pi}{6} \right)$$

$$= 32 \operatorname{cis} \left(\frac{\pi}{3} + \frac{\pi}{6} \right)$$

$$= 32 \operatorname{cis} \frac{\pi}{2}$$

$$= \underline{\underline{32i}}$$

$$zw = (4 + 4\sqrt{3}i)(2\sqrt{3} + 2i)$$

$$= 8\sqrt{3} + 8i + 24i + 8\sqrt{3}i^2$$

$$= 0 + 32i$$

$$b) \quad \frac{z}{w} = \frac{8 \operatorname{cis} \frac{\pi}{3}}{4 \operatorname{cis} \frac{\pi}{6}}$$

$$= \frac{8}{4} \operatorname{cis} \left(\frac{\pi}{3} - \frac{\pi}{6} \right)$$

$$= 2 \operatorname{cis} \frac{\pi}{6}$$

$$= 2 \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right)$$

$$= \underline{\underline{\sqrt{3} + i}}$$

$$\frac{z}{w} = \frac{4 + 4\sqrt{3}i}{2\sqrt{3} + 2i} = \frac{x(2 + 2\sqrt{3}i)}{x(\sqrt{3} + i)} \cdot \frac{\sqrt{3} - i}{\sqrt{3} - i}$$

$$= \frac{x(1 + \sqrt{3}i)(\sqrt{3} - i)}{x(2)}$$

$$= \frac{\sqrt{3} - i + 3i - \sqrt{3}i^2}{2}$$

$$= \frac{2\sqrt{3} + 2i}{2}$$

$$= \underline{\underline{\sqrt{3} + i}}$$

$$2. \quad (1 + \sqrt{3}i)^9$$

$$z = 2 \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$$

$$z = 2 \operatorname{cis} \frac{\pi}{3}$$

$$z^9 = 2^9 \operatorname{cis} \left(\frac{\pi}{3} \right)^9$$

$$z^9 = 512 \operatorname{cis} 3\pi \quad (\text{By De Moivre's Thm}).$$

$$z^9 = 512 \operatorname{cis} \pi$$

$$z^9 = 512(-1)$$

$$z^9 = \underline{\underline{-512}}$$

$$3. \quad z = (\cos \theta + i \sin \theta)$$

$$z^4 = (\cos \theta + i \sin \theta)^4 = \cos^4 \theta + 4 \cos^3 \theta i \sin \theta + 6 \cos^2 \theta i^2 \sin^2 \theta$$

$$+ 4 \cos \theta i^3 \sin^3 \theta + i^4 \sin^4 \theta$$

By De Moivre's Thm

$$\cos^4 \theta + i \sin^4 \theta = \cos^4 \theta + 4 \cos^3 \theta i \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4 \cos \theta i \sin^3 \theta + \sin^4 \theta$$

$$2. \quad a) \quad \cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta \quad (\text{Equating Real Parts})$$

$$b) \quad \sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta \quad (\text{Equating Imag. Parts})$$

$$1. \quad c) \quad \tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta} = \frac{4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta}{\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta}$$

÷ top and bottom by $\cos^4 \theta$

$$2. \quad \therefore \tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{\tan^4 \theta - 6 \tan^2 \theta + 1}$$

$$d) \quad \cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2$$

$$= \cos^4 \theta - 6 \cos^2 \theta + 6 \cos^4 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta$$

$$= \underline{\underline{8 \cos^4 \theta - 8 \cos^2 \theta + 1}}$$

$$e) \quad \text{By putting } x = \cos \theta$$

then the solutions to $8x^4 - 8x^2 + 1 = 0$

are the solutions to $\cos 4\theta = 0$

$$\text{i.e. } \cos 4\theta = 0 \Rightarrow 4\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

$$\cos 4\theta = \cos \frac{\pi}{2}, \cos \frac{3\pi}{2}, \cos \frac{5\pi}{2}, \cos \frac{7\pi}{2}$$

$$\Rightarrow \theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$$

$$4. \quad \text{Hence solutions to } 8x^4 - 8x^2 + 1 = 0$$

are $x = \cos \frac{\pi}{8}, x = \cos \frac{3\pi}{8}, x = \cos \frac{5\pi}{8}, x = \cos \frac{7\pi}{8}$.

$$f) \quad \text{Sum of Roots } -\frac{b}{a} = \frac{0}{8} = 0$$

$$\therefore \cos \frac{\pi}{8} + \cos \frac{3\pi}{8} + \cos \frac{5\pi}{8} + \cos \frac{7\pi}{8} = 0$$

$$g) \quad \text{Product of roots } \frac{c}{a} = \frac{1}{8}$$

$$\cos \frac{\pi}{8} \cdot \cos \frac{3\pi}{8} \cdot \cos \frac{5\pi}{8} \cdot \cos \frac{7\pi}{8} = \frac{1}{8}$$